

Aleyna Dilan Kiran

✉ Email  GitHub  LinkedIn  Google Scholar

Education

MSc in Computer Science 2023–Present

Universität des Saarlandes, Germany

Research Interests: Explainable Machine Learning, Multimodal VAEs, Bioinformatics¹

BEng in Bioengineering 2018–2022

Ege University, Turkey

CGPA: 3.53/4 (German equivalent: 1.7)

Thesis: Investigation of Celecoxib and COX-2 Interaction via Quantum Mechanical Methods

Research Experience

Machine Learning Group, Universität des Saarlandes, Germany

Thesis Student 06/2025–Present

- Conducting MSc thesis on Multimodal Variational Autoencoders (VAEs) with enhanced latent representation
- Funded by ELIZA — Konrad Zuse AI School (affiliated with ELLIS)

Research Assistant (HiWi) 10/2023–Present

- Currently developing a data-loading benchmark for Concept Bottleneck Models (CBMs)
- Benchmarked fairness datasets; developed reusable Jupyter notebooks
- Conducted data & model analysis for personalized deep-learning stress detection
- Co-authored an ICML workshop paper on personalized stress prediction
- Assisted in survey design on fairness, fair representation learning, and fairness adaptation for LLMs

Hochschule Ruhr West, Germany

ERASMUS+ Intern 03/2022–02/2023

- Evaluated workflow management systems (WFMS) in bioinformatics
- Assessed FAIRness (Findable, Accessible, Interoperable, Reusable) and security criteria
- Published a paper as first author; presented at symposiums

Molecular Modelling and Bioinformatics Lab, Ege University, Turkey

Project Leader / Researcher 06/2021–12/2022

- Led a fully funded research project
- Conducted quantum mechanical drug–amino acid interaction analysis
- Designed 3D molecular models; presented findings

Izmir Biomedicine and Genome Center (IBG), Turkey

Intern 10/2021–12/2021

- Neuro-Genomics Lab — Nanopore sequence analysis (minimap2, Samtools, SURVIVOR)
- Computational Structural Biology Lab — ML-based protein interaction analysis

Publications & Presentations

Publications

P3. Koyuncu, B., **Kiran, A.D.**, et al. (2024). Personalized Stress Prediction via Smartwatches. ICML ML4LMS Workshop.

P2. **Kiran, A.D.**, Can, M.C., Allmer, J. (2023). Evaluation Criteria for Workflow Management Systems. Fortune Journals.

P1. Kilpatrick, A.M., ..., **Kiran, A.D.**, et al. (2022). Domain-Specific Open Educational Resources. Bioinformatics.

Poster Presentations

PP2. **Kiran, A.D.**, Can, M.C., Allmer, J. (2022). Criteria for Workflow Management Systems. 15th Int. Symposium on Health Informatics and Bioinformatics.

PP1. **Kiran, A.D.**, Selçuki, C. (2022). Investigation of Celecoxib Molecule and COX-2 Interaction. 15th Int. Symposium on Health Informatics and Bioinformatics.

Oral Presentation

OP1. **Kiran, A.D.**, Can, M.C., Allmer, J. (2022). Criteria for the Evaluation of Workflow Management Systems for Scientific Data Analysis. Workshop Medical OMICS, Germany.

¹Previously enrolled in an MSc in Bioinformatics (April 2023 – September 2024); transferred to the MSc in Computer Science.

Teaching

Universität des Saarlandes, Germany

Tutor

04/2026–Present

- Currently leading weekly tutorial sessions for the ‘Machine Learning’ course
- Assisting with midterm preparation and grading
- Supporting students with coursework and exam preparation

Technical Skills

Programming: Python, C++, R

Frameworks: PyTorch, TensorFlow, scikit-learn

ML: Deep Learning, Probabilistic Machine Learning, Transformers, Variational Autoencoders

Tools: Git, Linux, LaTeX

Awards & Scholarships

ELIZA Zuse School Scholarship	2024
Graduated in Top 5% (Ranked 4th)	2022
ERASMUS+ Internship Fellowship	2022
TÜBİTAK Research Grants (Bachelor’s thesis and internship)	2021

Extracurricular Activities

ELLIS Lisbon Machine Learning School (LxMLS)	2025
ISCB RSG Turkey — Translation Team Leader	2024
ISCB RSG Turkey — Website Team Leader	2023
ISCB Student Symposium – Organizing Committee Member	2022
15th International Symposium on Health Informatics and Bioinformatics	2022
Workshop Medical OMICS	2022
ISCB SCS Wikipedia Hackathon	2021

Languages

Turkish (Native), English (Fluent), German (Intermediate)